

Development of Low-Cost, Low-Power Consumption, and Narrow-Frame IGZO Panels for IT (MFD&DeMUX)

Tatsuya Kawasaki, Masafumi Sugino, Yohei Takeuchi, Kengo Hara, Kohji Saitoh, Satoshi Horiuchi, Daiji Kitagawa, Tetsuo Kikuchi, Hajime Imai, Tohru Daitoh

Sharp Display Technology Corporation, Mie, Japan

Abstract

We are developing narrow-bordered panels that are low cost and consume less power. We are also developing MFD technology with the aim of reducing practical power consumption. While conventional displays use the same refresh rate for the entire screen, MFD technology allows images to be displayed at different refresh rates. This makes it possible to reduce power consumption while still displaying moving images. We are also developing DeMUX technology for BCE structure IGZO-TFTs. This technology makes it possible to reduce the number of source drivers implemented on the panel, thereby reducing panel costs. Furthermore, by combining DeMUX technology with Single Gate, it is possible to suppress the increase in the width of the left and right borders. By combining these two technologies, we can realize low-cost, low-power, narrow-bordered panels.

Author keywords

IGZO; Oxide TFT; BCE; DeMUX; narrow-bordered; MFD; Dual Gate; low power consumption; low cost, Source driver reduction

1. Introduction

Displays are made up of many components. IC chips (source drivers) are particularly expensive and in high demand. IC chips are used in a variety of products, not just displays, and the previous global semiconductor shortage had a major impact on other industries as well. Although the supply is currently expected to improve, there are concerns that future advances in IT technology and the global environment could lead to a supply shortage. To address these issues, development efforts are underway to reduce the number of source drivers used on panels. Dual Gate and De-multiplexer (DeMUX) technologies are available to reduce the number of source lines. Dual Gate is currently in production, and DeMUX is being produced using LTPS. IGZO-DeMUX is currently under development.[1][2] Using these technologies will not only stabilize supply but also reduce panel costs. Furthermore, future demand for displays is expected to increase for laptops and tablets. This trend is evident in both LCD and OLED. Along with stable supply, design and portability are also important. There is a demand for displays with narrow-bordered and low power consumption that are easy to carry. In recent years, the use of streaming services for movies, sports, and other content, along with communication tools, has increased. This increase necessitates measures to reduce power consumption while maintaining the display performance of these videos.

To reduce power consumption, a function is previously available to change the refresh rate from 1 to 120 Hz across the entire screen as

needed. In standby mode, the refresh rate can be lowered to reduce power consumption. However, when displaying video, a high refresh rate is required for the entire screen, resulting in increased power consumption. Therefore, we have developed Multi-Frequency Display (MFD) technology, which allows the refresh rate of only a portion of the screen to be changed. Here, we report that this technology enables power consumption reduction even while displaying video in only a portion of the screen.

2. MFD technology

MFD technology changes the refresh rate of a certain area of the display screen. The display is divided into video and still image areas, with the video data that changes frequently transmitted at a high refresh rate. In contrast, the still image data can be separated and transmitted at a low refresh rate.

Figures 1a and 1b show examples of using MFD. Figure 1a shows an example when displaying video. The area in the center of the screen where the video is displayed is scanned at a high refresh rate, while the rest of the screen is scanned at a low refresh rate. This makes it possible to maintain the quality of the video display while reducing power consumption in the still image areas. MFD technology is not limited to video display situations; it is also effective in office typing situations, as shown in Figure 1b.

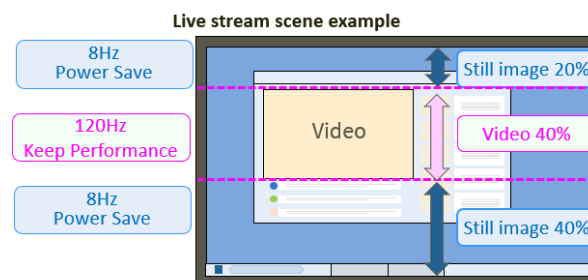


Figure 1a. MFD display example1

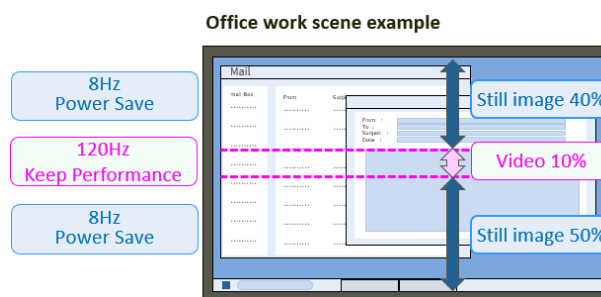


Figure 1b. MFD display example2

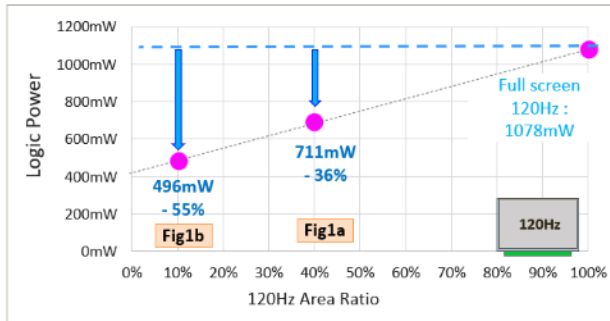


Figure 2. Logic Power

Figure 2 shows the power consumption of a panel that applies MFD technology. In the case of Figure 1a, the high refresh rate area accounts for 40% of the entire screen. Compared to when the entire screen is operated at 120Hz, power consumption can be reduced by 36%. In the case of Figure 1b, the high refresh rate area accounts for 10% of the entire screen. This reduces power consumption by 55%.

This section explains the design of the gate-on-array (GOA) circuit, which controls the driving of gate lines. Figure 3a shows a schematic diagram of an existing GOA circuit. The existing circuit uses a GCK signal to drive the GCK circuit at 120 Hz, sequentially outputting the gate lines. Figure 3b shows a GOA circuit using MFD. The MFD adds a new QCK circuit to the GCK circuit. The QCK circuit operates each gate stage at 120 Hz to charge the GCK circuit's netAG. The GCK circuit receives a 120 Hz GCK signal for the video display stage, but an 8 Hz GCK signal for the still image stage. When no gate signal is output, the GCK signal is also paused. Because the source signal is simultaneously paused during this period, it is possible to shut down the source amplifier circuit in the source driver and reduce power consumption. However, a GOA circuit for an MFD requires wiring for the QCK signal and a control circuit, which increases the circuit size.

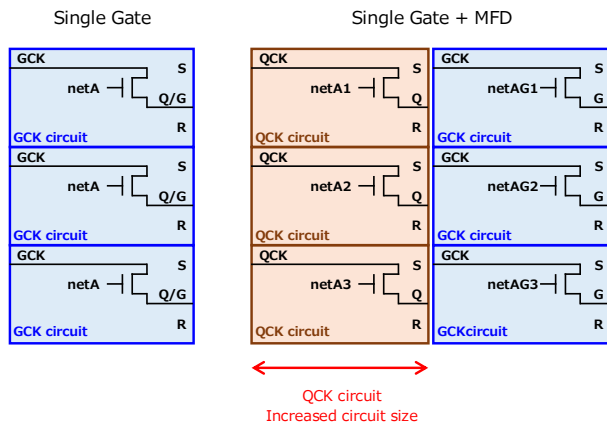


Figure 3a.
Existing circuit schematic

Figure 3b.
MFD circuit schematic

3. Combining Source Driver Reduction Technology with MFD

3.1 Dual Gate Technology

Dual Gate (DG) technology is one way to reduce the number of source drivers. This technology doubles the number of gate lines,

halving the number of source lines required. Figure 4a shows schematic diagrams of Single Gate (SG) and DG pixel layouts. SG connects one gate line and one source line to each pixel. DG connects one source line and two gate lines to two pixels.

Figure 4b explains the details of DG. A single source line connects to two adjacent pixels. One pixel is connected to the upper gate line and the other to the lower gate line. Although the source signal is shared, each pixel is driven by changing the timing at which the on-signal is input to the gate line. Figure 4c shows the relationship between SG/DG technology and source drivers, taking a model requiring six source drivers with SG as an example. DG reduces the number of source lines to half that of SG, making it possible to reduce the number of source drivers to three. However, doubling the number of gate lines halves the vertical pitch between gate lines. As the circuit area per line shrinks, it needs to be expanded horizontally. In addition, the driving time per gate line needs to be shortened, which necessitates an increase in the size of the transistor. As a result, the border width on the left and right sides increases. As for pixels, an increase in the number of gate lines reduces the area of the pixel's transparent region, which reduces transmittance.

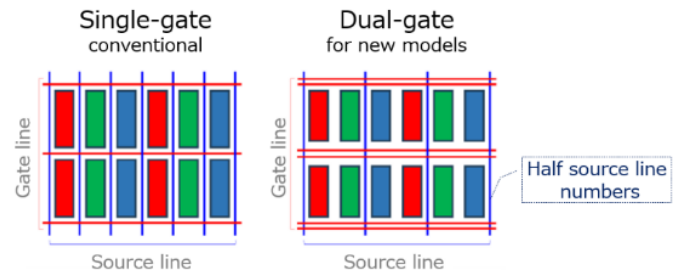


Figure 4a. SG and DG pixel layout

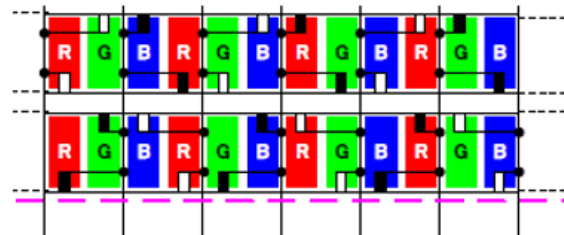


Figure 4b. Pixel layout of DG



Figure 4c. Relationship between SG/DG and source driver

3.2 DeMUX Technology

Another method for reducing the number of source drivers is DeMUX technology. Figure 5 shows a schematic diagram of this technology. A DeMUX circuit that distributes signals is installed at the base of the source lines, and the signals from the source driver are divided and distributed by time. This reduces the number of source wiring that supplies the source signals, making it possible to reduce the number of source drivers.

However, because the signals are divided by time, it is necessary to charge the source bus lines in a short period of time. For this reason, DeMUX transistors are required to have a high ON-current capability. We implement this technology using IGZO transistors with the back-channel etch (BCE) structure. The number of gate wiring in a display area using DeMUX is the same as that of SG, and there is no reduction in transmittance or expansion of GOA circuit width unlike with DG.

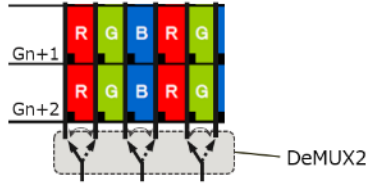


Figure 5. DeMUX schematic diagram

3.3 Comparison of DG and DeMUX

Table 1 shows the performance comparison results for the two technologies. For both DG and DeMUX technologies, the number of source drivers can be reduced from six to three. The left and right border areas are primarily for the GOA circuit and routing wiring, and the seal structure area. The seal structure area is necessary to ensure panel strength. SG border widths are generally 2 mm, depending on the specifications. While the DG circuit area is larger horizontally, sharing it with the seal structure area allows for the same border width as SG. DG increases the number of gate lines in the pixel area, reducing transmittance. This increases backlight power. DG increases the number of gate lines, increasing GOA circuit power consumption. However, because the reduced number of sources reduces the power related to source signals, logic power is equivalent to that of SG. DeMUX's GOA circuit is the same as SG's, and the left and right border areas are the same as SG's. DeMUX's pixel area also has the same structure as SG's, and its backlight power is also the same as SG's. DeMUX requires the addition of a DeMUX circuit to the source line, and the DeMUX circuit must operate at high speed, which increases logic power. However, it is possible to suppress the increase in power consumption when using low refresh rate drive. When using low refresh rate drive, SG consumes the least power, followed by DeMUX. DG consumes the most power.

Table 1. SG/DG/DeMUX comparison chart

14inch / QHD+ 400nit / sRGB 100% w/o Touch integrated display w/o MFD		Single Gate	Dual Gate	DeMUX 2
Source Driver numbers		6	3	3
Border L/R		2.0mm	2.0mm	2.0mm
Transmittance		100%	94% ※6% lower than SG	100%
Logic Power	120Hz 1Hz	0.73W max 0.27W max	0.73W max 0.27W max	0.92W max 0.28W max
BL power	400nit	2.81W max	2.98W max	2.81W max
Total power	120Hz / 400nit 1Hz / 400nit	3.54W max 3.08W max	3.71W max 3.25W max	3.73W max 3.09W max

3.4 Combining source driver reduction technology with MFD

Table 2 shows the survey results for the left and right border widths. The left and right border widths for DG only and SG+MFD are 2mm, the same as for SG only. Although both DG and MFD require larger circuit areas, using either technology does not affect the width of the left and right borders, because they share the free space under the seal area.

On the other hand, when DG is combined with an MFD, the circuit area for both the DG and MFD increases. Because it is not possible to accommodate all the circuits in the area under the seal, the width of the left and right borders increases to 2.5 mm. While typical models have left and right border widths of around 2.5mm, high-end narrow-bordered models often require border widths of less than 2mm. Therefore, combining DG and MFD technologies makes it difficult to meet the narrow-bordered requirements of high-end models.

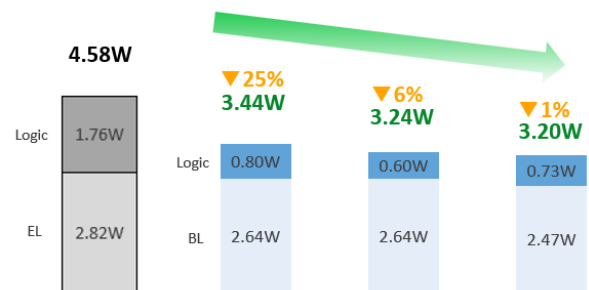
In contrast, DeMUX technology uses SG structure, so even in combination with MFD technology, the left and right border widths are 2mm. Therefore, DeMUX technology is considered advantageous when combined with MFD technology, which has a large GOA circuit size.

Table 2. Left and right border widths when combining source driver reduction technology with MFD

14inch / QHD+			
Technology combination	SG	DG	DeMUX
w/o MFD	2.0mm	2.0mm	2.0mm
w/ MFD	2.0mm	2.5mm	2.0mm

Figure 6 shows the results of module power consumption. When driving OLED and LCD without MFD at 120Hz, DG LCD can reduce power consumption by approximately 25% compared to OLED panel. Adding MFD technology to this reduces power consumption by an additional 6%. With DeMUX and MFD, a further reduction of 1% is possible.

Comparison at 120Hz



	2025	2025	2026	2027
14" 2880x1800				
Tandem OLED	LCD			
Other companies' products	Now	+MFD	+DeMUX	
OPR 72%	Dual Gate	←	DeMUX2	
Full screen 120Hz	←	MFD 8Hz 30% 120Hz 40% 8Hz 30%	←	
400nit				
DCI-P3 100%				

Figure 6. Module power consumption comparison

We prototyped a panel equipped with DeMUX and MFD for PC. Fig.7 shows the specifications and the actual panel. It is lighting up normally and images are displayed clearly. We are currently evaluating the prototype, and it is scheduled to be completed in March.

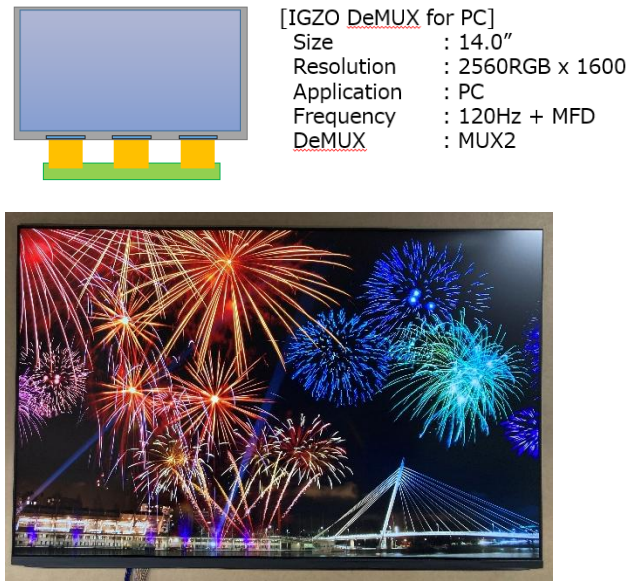


Figure 7. Specifications and display images of a panel equipped with DeMUX and MFD

4. Conclusion

We are developing low-cost, low-power, narrow-bordered panels by combining MFD and DeMUX technology. IGZO DeMUX technology has enabled us to develop a panel with fewer source drivers at a low cost. Simulation results have shown that combining MFD and DeMUX reduces power consumption. We have also succeeded in designing a border width of less than 2 mm on both sides. We prototyped a panel equipped with DeMUX and MFD for PC. We are currently evaluating the prototype, and it is scheduled to be completed in March

5. References

- [1] K. Hara, Y. Takeuchi, M. Sugino, T. Kawasaki, H. Imai, T. Daitoh SID 79.3 1083-1086(2025)
- [2] Y. Takeuchi, A. Hachiya, M. Sugino, K. Hara, T. Kawasaki, H. Imai, SID P78(2024)